CO1035: Game Concepts
Math Exercises – Vectors

Exercises

1. If **P1** = (50, 50, 50), **P2** = (-11, 22, -33), **A** = (3, 4, 5), **B** =(5, 7, 9), **C** = (10, 20), **D** = (34, 50), calculate the following:
   - **a.** P1P2, i.e., the vector between P1 and P2.
   - **b.** P2P1, i.e., the vector between P2 and P1.
   - **c.** AB, i.e., the vector between A and B.
   - **d.** DC, i.e., the vector between D and C.
   - **e.** BC, i.e., the vector between B and C.

2. If **a** = (1,2,3), **b** = (5,6,7), **c** = (5,10), **d** = (0,13,8) and **e** = (4,9), calculate the following:
   - **a.** a + b
   - **b.** b – a
   - **c.** 12b
   - **d.** 4d
   - **e.** 5a – b
   - **f.** e + 2c
   - **g.** $\|c\|$
   - **h.** $\|b\|$
   - **i.** $\hat{a}$
   - **j.** $\hat{e}$

Solutions:

1a) P1P2, i.e. the vector between P1 and P2
P1 = (50, 50, 50)
P2 = (-11, 22, -33)
P1P2 = P2 - P1 = (P2x – P1x, P2y – P1y, P2z – P1z)
P1P2 = (-11 – 50, 22 – 50, -33 -50)
P1P2 = (-61, -28, -83)

1b) P2P1, i.e. the vector between P2 and P1
P1 = (50, 50, 50)
P2 = (-11, 22, -33)
P2P1 = P1 – P2 = (P1x – P2x, P1y – P2y, P1z – P2z)
P1P2 = (50 – (-11), 50 - 22, 50 – (-33))
P1P2 = (61, 28, 83)

1c) AB, i.e. the vector between A and B
A = (3, 4, 5)
B = (5, 7, 9)
AB = B – A = (Bx – Ax, By – Ay, Bz – Az)
AB = (5 – 3, 7 – 4, 9 – 5)
AB = (2, 3, 4)

1d) DC, i.e. the vector between D and C
C = (10, 20)
D = (34, 50)
DC = C – D = (Cx – Dx, Cy -Dy, Cz – Dz)
DC = (10 – 34, 20 -50, 0 - 0)
DC = (-24, -30) or (-24, -30, 0)

1e) BC, i.e. the vector between B and C
B =(5, 7, 9)
C = (10, 20)
BC = C – B = (Cx – Bx, Cy – By, Cz – By)
BC = (10 – 5, 20 – 7, 0 – 9)
BC = (5, 13, -9)

2a) a + b
a = (1,2,3)
b = (5,6,7)

a + b = (ax + bx, ay + by, az + bz)
a + b = (1 + 5, 2 +6, 3 +7)
a + b = (6, 8, 10)

2b) b – a
a = (1,2,3)
b = (5,6,7)

b – a = (bx – ax, by – ay, bz – az) b – a = (5 – 1, 6 – 2, 7 – 3)
b – a = (4, 4, 4)

2c)

$\frac{1}{2}$b
b = (5, 6, 7)
$\frac{1}{2}b = \frac{1}{2}(5,6,7)$
$\frac{1}{2}b = \left(\frac{5}{2},\frac{6}{2},\frac{7}{2}\right)$
$\mathbf{\frac{1}{2}b = (2.5, 3, 3.5)}$

2d) 4d
d = (0, 13, 8)

4d = 4(0, 13, 8)
4d = (4×0, 4×13, 4×8)
4d = (4, 52, 32)

2e) 5a – b
a = (1,2,3)
b = (5,6,7)

5a – b = 5(1,2,3) – (5,6,7)
5a – b = (5, 10, 15) – (5, 6, 7)
5a – b = (5 – 5, 10 – 6, 15 – 7)
5a – b = (0, 4, 8)

2f) e + 2c
c = (5,10)
e = (4,9)

e + 2c = (4, 9) + 2(5, 10)
e + 2c = (4, 9) + (10, 20)
e+ 2c = (4+10, 9+20)
e + 2c = (14, 29)

2g)

$\|c\|$

c = (5,10)

$\|c\| = \sqrt{cx^2 + cy^2 + cz^2}$

$\|c\| = \sqrt{5^2 + 10^2 + 0^2}$

$\boldsymbol{\|c\| = \sqrt{125} = 5\sqrt{5}}$

2h)

$\|b\|$

b = (5, 6, 7)

$\|b\| = \sqrt{bx^2 + by^2 + bz^2}$

$\|b\| = \sqrt{5^2 + 6^2 + 7^2}$

$\boldsymbol{\|b\| = \sqrt{110}}$

2j)

$\hat{e}$

e = (4,9),

$\hat{e} = \left(\frac{ex}{\|e\|}, \frac{ey}{\|e\|}, \frac{ez}{\|e\|}\right) =$

$\|e\| = \sqrt{ex^2 + ey^2 + ez^2}$

$\|e\| = \sqrt{4^2 + 9^2 + 0^2}$

$\|e\| = \sqrt{97}$

$\hat{e} = \left(\frac{ex}{\|e\|}, \frac{ey}{\|e\|}, \frac{ez}{\|e\|}\right) = \left(\frac{4}{\sqrt{97}}, \frac{9}{\sqrt{97}}\right)$